

Ch 6: C: 1a, b, c, f
2a, b, d
4, 5, 6

(C1) For $u = (x_1, y_1), v = (x_2, y_2) \in \mathbb{R}^2$, $T(x, y) = (x+y, x)$

$$\begin{aligned} a) T(u) + T(v) &= (x_1 + y_1, x_1) + (x_2 + y_2, x_2) \\ &= (x_1 + x_2 + y_1 + y_2, x_1 + x_2) \\ &= T(u+v) \end{aligned}$$

for $a \in \mathbb{R}$,

$$\begin{aligned} T(au) &= (a(x_1 + y_1), ax_1) \\ &= a(x_1 + y_1, x_1) = aT(u) \end{aligned}$$

$\Rightarrow$ T is
a linear
map.

b) T is surjective if $\text{range}(T) = \mathbb{R}^2$

i.e. for any $y \in \mathbb{R}^2$ there is an $x \in \mathbb{R}^2$ such that $T(x) = y$.

Let $(y_1, y_2) \in \mathbb{R}^2$. $T(x, x_2) = (y_1, y_2)$ for $x_1 = y_2, x_2 = y_1 - y_2$

for any $(y_1, y_2) \in \mathbb{R}^2$, $T(y_2, y_1 - y_2) = (y_1, y_2)$

Thus T is surjective.

c) The only solution to $T(x, y) = (x+y, x) = (0, 0)$ is $(x, y) = (0, 0)$
 $= 0 \in \mathbb{R}^2$, so $\text{null}(T) = \{0\}$ which has dimension $\boxed{0}$.

f) For $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $F(x, y) = (x+y, x+1)$

Let $a \in \mathbb{R}$, $u = (x, y)$

$$F(au) = F(ax, ay) = (ax+ay, ax+1)$$

$$\neq (a(x+y), a(x+1)) = aF(u)$$

so this map is not linear

(C2) Define $T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ -x \end{bmatrix}$

a) for any $\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$,

$$T\begin{bmatrix} -y \\ x \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \text{ so } T \text{ is surjective.}$$

b) $T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ if and only if $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$
equivalently: for $x \in \mathbb{R}^2$, $Tx = 0 \Leftrightarrow x = 0$
Null space is $\{0\}$ and $\dim(\{0\}) = \boxed{0}$.

d) For $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $F(x, y) = (x+y, x+1)$

Note that $0 \notin \text{null}(F)$

$$\text{null}(F) = \{(-1, 1)\}$$

$$F(-1, 1) = 0 \text{ but } F(0, 0) = (0, 1) \neq 0 \in \mathbb{R}^2$$

Then for $x \in \mathbb{R}^2$,

$$F(x) = F(x+0) \neq F(x) + F(0) !$$

So F is not linear

Math 67 – HW 4 Solutions

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Chapter 6

C4. Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ to be $f(x, y) = (x^3 + y^3)^{1/3}$.

C5. A linear map $T : \mathbb{F}^4 \rightarrow \mathbb{F}^2$ with a 2-dimensional nullspace. We show that it has to be surjective, i.e., $\text{range}(T) = \mathbb{F}^2$.

By the Dimension Formula (Theorem 6.5.1)

$$\dim(\mathbb{F}^4) = \dim(\text{range}(T)) + \dim(\text{null}(T)).$$

So $\dim(\text{range}(T)) = 2$. We know that $\text{range}(T) \subseteq \mathbb{F}^2$ and since $\dim(\text{range}(T)) = \dim(\mathbb{F}^2)$, it follows that $\text{range}(T) = \mathbb{F}^2$. (This was the result from chap 5 proof problem 4, which was on HW#3).

C6. No linear map $T : \mathbb{F}^5 \rightarrow \mathbb{F}^2$ can have a 2-dimensional nullspace:

It follows from the dimension formula that $\text{range}(T) = 3$, but this is impossible because $\text{range}(T) \subseteq \mathbb{F}^2$, which has dimension 2. (by Theorem 5.4.4 part 1)

P1. Let V and W be vector spaces over $\mathbb{F}$ with V finite-dimensional, and let U be any subspace of V . Given a linear map $S \in \mathcal{L}(U, W)$, there exists a $T \in \mathcal{L}(V, W)$ such that for every $u \in U$, $S(u) = T(u)$.

Proof. The vector space V is finite dimensional so let $n = \dim(V)$ and let $m = \dim(U)$, where $m \leq n$ by Theorem 5.4.4 since $U \subseteq V$. By Corollary 5.3.6 there exists a basis $\{u_1, u_2, \dots, u_m\}$ of U . Note that each vector $u_j \in V$ also, and the set $\{u_j\}$ is linearly independent. By Theorem 5.3.7 (Basis Extension Theorem) there exist $n - m$ more vectors $v_j \in V$ such that

$$\{u_1, u_2, \dots, u_m, v_1, v_2, \dots, v_{n-m}\} \text{ is a basis for } V. \quad (*)$$

Now, we define the set $\{w_1, w_2, \dots, w_n\}$ as follows: let $w_j = S(u_j)$ for $j = 1, 2, \dots, m$, and for $j = m + 1, m + 2, \dots, n$, let w_j be an arbitrary vector in W .

By Theorem 6.1.3 there exists a linear map $T \in \mathcal{L}(V, W)$ such that the vectors in $(*)$ map to $\{w_j\}_{j=1}^n$. More specifically, $T(u_j) = w_j$ for $j = 1, 2, \dots, m$, and $T(v_j) = w_{m+j}$ for $j = 1, 2, \dots, n - m$.

Since $w_j = S(u_j) = T(u_j)$ for $j = 1, 2, \dots, m$ and the set $\{u_j\}_{j=1}^m$ is a basis for U , for any $u \in U$ there exist some $a_1, a_2, \dots, a_m \in \mathbb{F}$ so that

$$\begin{aligned} S(u) &= S(a_1 u_1 + a_2 u_2 + \dots + a_m u_m) \\ &= a_1 S(u_1) + a_2 S(u_2) + \dots + a_m S(u_m) \\ &= a_1 T(u_1) + a_2 T(u_2) + \dots + a_m T(u_m) \\ &= T(a_1 u_1 + a_2 u_2 + \dots + a_m u_m) \\ &= T(u) \end{aligned}$$

Note: In summary, what we did was take a basis for U (which is a subspace of V), and extend it to a basis of V . Then, we showed that we can take the map $S : U \rightarrow W$, and extend it into a map from the bigger space V into W .

The point of this is to show that we only need to define a map on a smaller subspace, and it can be extended to a map on the whole space. This is an important result in functional analysis called the Hahn-Banach theorem.

P2. Let V and W be vector spaces over $\mathbb{F}$ and $T \in \mathcal{L}(V, W)$ is injective. Given a linearly independent set of vectors $\{v_1, v_2, \dots, v_n\} \subset V$, the set $\{T(v_1), T(v_2), \dots, T(v_n)\} \subset W$ is also linearly independent.

Proof. Consider the homogeneous linear combination

$$a_1 T(v_1) + a_2 T(v_2) + \dots + a_n T(v_n) = 0, \quad (*)$$

where $a_1, a_2, \dots, a_n \in \mathbb{F}$ are some scalars whose values are to be determined. Linearity of T implies that

$$T(a_1 v_1 + a_2 v_2 + \dots + a_n v_n) = 0.$$

Since T is injective, it follows from proposition 6.2.6 (T maps only 0 to 0) that

$$a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0.$$

The set $\{v_1, v_2, \dots, v_n\}$ is linearly independent, so we must have that

$$a_1 = a_2 = \dots = a_n = 0. \quad (**)$$

Since $(*)$ implies $(**)$, the set $\{T(v_1), T(v_2), \dots, T(v_n)\}$ must be linearly independent. $\square$

Note: If it is clear from the context how many vectors there are and which ones you are referring to, it is ok to use the shorthand $\{v_j\}$ for $\{v_j\}_{j=1}^n = \{v_1, v_2, \dots, v_n\}$. We can also do this with sums $\sum a_j v_j = a_1 v_1 + a_2 v_2 + \dots + a_n v_n$. Then, we can say that by linearity of T ,

$$T\left(\sum a_j v_j\right) = \sum a_j T(v_j)$$

which is easier to write.

P3. Let U, V , and W be vector spaces over $\mathbb{F}$, and let $S \in \mathcal{L}(U, V)$ and $T \in \mathcal{L}(V, W)$ be injective linear maps. Then their composition $T \circ S$ is injective.

Proof. We will show that $\text{null}(T \circ S) = \{0\}$, from which it follows by Proposition 6.2.6 that $T \circ S$ is injective.

Let $x \in \text{null}(T \circ S)$. Then $T(S(x)) = 0$. Since T is injective, Prop. 6.2.6 implies that $S(x) = 0$. Likewise, since S is injective, it follows that $x = 0$. Then the only member of $\text{null}(T \circ S)$ is 0. $\square$

P7. Let U, V , and W be finite-dimensional vector spaces over $\mathbb{F}$ with $S \in \mathcal{L}(U, V)$ and $T \in \mathcal{L}(V, W)$. Then

$$\dim(\text{null}(T \circ S)) \leq \dim(\text{null}(T)) + \dim(\text{null}(S))$$

Proof. The nullspace of a finite dimensional vector space is a subspace, and also finite dimensional, so for $n = \dim(\text{null}(S))$ and $m = \dim(\text{null}(T))$, let $\{u_1^{(1)}, u_2^{(1)}, \dots, u_n^{(1)}\} \subset U$ be a basis for $\text{null}(S)$ and let $\{v_1, v_2, \dots, v_m\} \subset V$ be a basis for $\text{null}(T)$. Some subset of the basis vectors $\{v_j\}$ are in the range of S . Without loss of generality let's call them $\{v_1, v_2, \dots, v_\ell\}$ where $\ell \leq m$. Let's denote their pre-images under S to be $\{u_1^{(2)}, u_2^{(2)}, \dots, u_\ell^{(2)}\}$, so that $v_j = S(u_j^{(2)})$ for $j = 1 \dots \ell$. Then the set

$$\{u_1^{(1)}, u_2^{(1)}, \dots, u_n^{(1)}, u_1^{(2)}, u_2^{(2)}, \dots, u_\ell^{(2)}\} \quad (\star)$$

is a basis for $\text{null}(T \circ S)$, and consists of $n + \ell \leq n + m$ basis vectors. $\square$

Note: how do we know that $(\star)$ is linearly independent? Because $\text{span}\{u_j^{(1)}\} = \text{null}(S)$ and $u_j^{(2)} \notin \text{null}(S)$ for any j .